SSA Solar Potential – Summary Statistics Report

Data Source: NASA POWER (Daily GHI, 2024–2026)

Cities Analysed: 7 SSA Urban Centers

Metric: Global Horizontal Irradiance (GHI, kWh/m²/day)

1. Regional Solar Potential Overview

Region	Mean GHI	Median GHI	Std Dev	Min	Max
East Africa	5.36	5.43	1.21	0.66	8.10
Southern Africa	5.32	5.12	1.55	0.91	9.38
West Africa	4.68	4.93	1.05	0.27	6.87

2. Top Opportunity Cities

Rank	City	Mean GHI	Std Dev
1	Nairobi, Kenya	5.78	1.18
2	Dar es Salaam, Tanzania	5.66	1.31
3	Johannesburg, South Africa	5.32	1.55
4	Kampala, Uganda	5.11	1.01
5	Kigali, Rwanda	4.88	1.08
6	Accra, Ghana	4.77	0.94
7	Lagos, Nigeria	4.59	1.14

3. Executive Insight

East Africa demonstrates the strongest solar opportunity across SSA, led by Nairobi and Dar es Salaam. While Southern Africa shows high peak irradiance, East Africa combines strong averages with relative stability, making it the most attractive region for solar adoption under the Model 2 framework.